

ONBRD-06: Organizing Your Environment

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Tags, Segments, and Naming Conventions

As your Dynatrace environment grows, organization becomes critical. This notebook covers how to structure your environment with tags, segments, and naming conventions for maintainability and access control.

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Prerequisites

- Admin or Configurator access to Dynatrace
- Entities discovered (hosts, services)
- Understanding of your organizational structure

OneAgent Attribute Enrichment (OneAgent 1.331+)

Requires: OneAgent version **1.331** or later

OneAgent can enrich **all telemetry signals** (metrics, spans, logs, events, entities) with custom metadata at the source — before data reaches the Dynatrace platform. This is more efficient than server-side tagging (auto-tags) because enrichment happens on the host and propagates to all Smartscape nodes.

Primary Fields (standardized from Semantic Dictionary):

- `dt.security_context` — data governance and access control
- `dt.cost.costcenter` — cost allocation
- `dt.cost.product` — product attribution

Primary Tags (custom key-value pairs):

- `primary_tags.environment` — environment identification (production, staging, etc.)
- `primary_tags.team` — team ownership
- `primary_tags.business_unit` — organizational unit

Configuration:

```
# Set during OneAgent installation
Dynatrace-OneAgent-Linux.sh --set-host-
tag="primary_tags.environment=production" --set-host-
tag="dt.security_context=confidential"

# Set on existing agents via oneagentctl
oneagentctl --set-host-tag="primary_tags.environment=production"
oneagentctl --set-host-tag="dt.cost.costcenter=12345"

# Per-process via environment variable (overrides host-level)
DT_TAGS="primary_tags.team=platform primary_tags.environment=production"
```

Benefits over auto-tagging:

Aspect	Auto-Tags (Server-Side)	Attribute Enrichment (Agent-Side)
-----	-----	-----
When applied	After data arrives at platform	At the source, before transmission
Scope	Entity tags only	All signals: metrics, spans, logs, events, entities
Grail integration	Limited	Full — feeds OpenPipeline routing, bucket assignment, permissions

| Cost allocation | Not supported | `dt.cost.costcenter` , `dt.cost.product` fields |
| Security context | Not supported | `dt.security_context` for data governance |

See: [Primary Grail fields and tags enrichment through OneAgent](#)

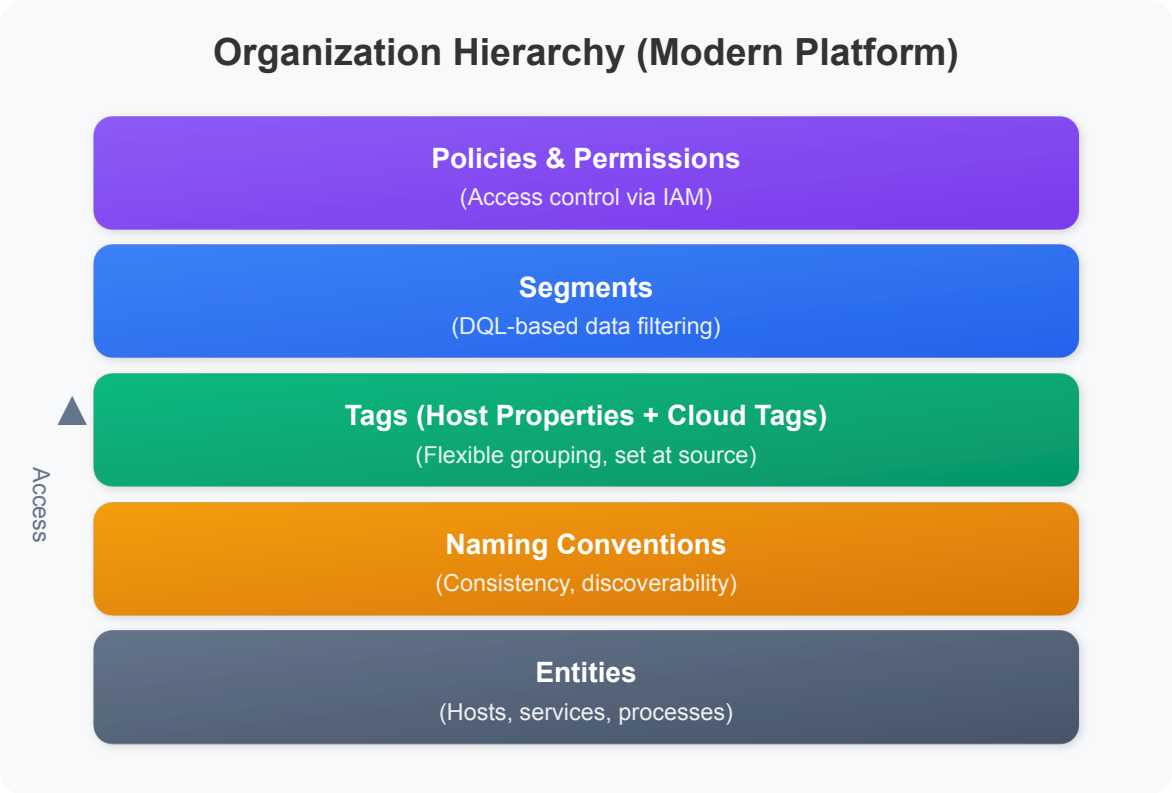
Version Support Policy: OneAgent and ActiveGate versions are supported for **9 months (Standard)** or **12 months (Enterprise)**. Third-party technologies are supported for 6 months beyond vendor EOL. See [Support Policy](#).

1. Why Organization Matters

Without organization, Dynatrace environments become difficult to manage:

Problem	Impact
No structure	Can't find entities quickly
No ownership	Don't know who to contact
No access control	Everyone sees everything
No filtering	Dashboards show irrelevant data
No grouping	Can't compare similar systems

Modern Organization Building Blocks



Note: The modern Dynatrace platform uses **Segments** for data filtering and **Policies** for access control. This replaces the legacy Management Zones approach.

2. Modern Organization Building Blocks

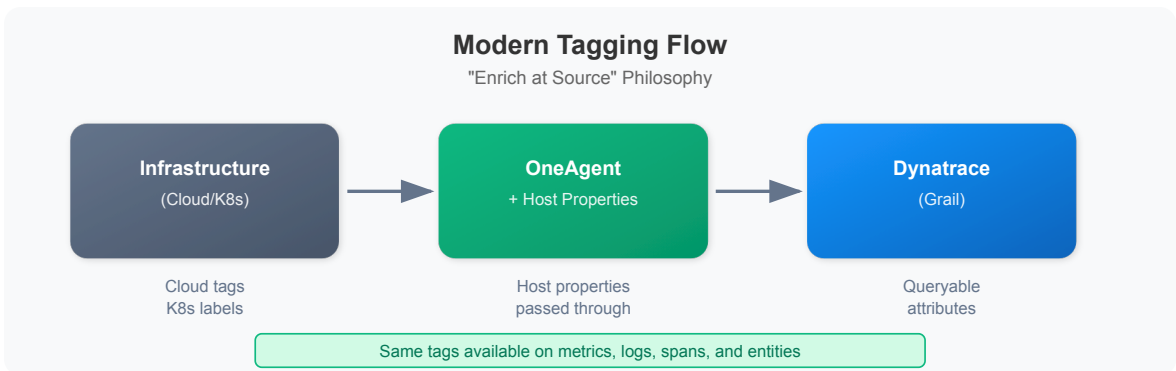
The modern Dynatrace platform (Gen3/Grail) uses a "tag at source" approach rather than rule-based auto-tagging:

Tag Sources

Source	How It Works	Best For
Host Properties	Set via OneAgent configuration	Environment, team, tier metadata

Source	How It Works	Best For
Cloud Provider Tags	Automatically imported from AWS/Azure/GCP	Cloud resource organization
Kubernetes Labels	Imported from K8s metadata	Container workload organization
OpenTelemetry Attributes	Set in instrumentation	Custom service attributes

The "Enrich at Source" Philosophy



Why "Tag at Source"?

Benefit	Description
Consistent	Same tags on metrics, logs, spans, and entities
Scalable	No processing overhead to apply rules
Traceable	Tags come from the source of truth
Real-time	No delay waiting for rule evaluation

3. Tagging with Host Properties and Cloud Tags

Host Properties (OneAgent)

Set custom properties during OneAgent installation or via configuration:

During Installation:

```
# Linux
sudo /bin/sh Dynatrace-OneAgent.sh \
  --set-host-property=env=production \
  --set-host-property=team=platform \
  --set-host-property=tier=backend

# Windows
.\Dynatrace-OneAgent.exe --set-host-property=env=production --set-host-property=team=checkout
```

Via Configuration File:

Note: The configuration file path depends on your OneAgent version. Check your version with `oneagentctl --version` to determine the correct path.

OneAgent Version	Configuration File Path
< 1.225	/var/lib/dynatrace/oneagent/agent/config/hostcustomproperties.conf
≥ 1.225	/var/lib/dynatrace/oneagent/agent/config/custom.properties

```
# OneAgent < 1.225:
/var/lib/dynatrace/oneagent/agent/config/hostcustomproperties.conf
# OneAgent ≥ 1.225: /var/lib/dynatrace/oneagent/agent/config/custom.properties
env=production
team=platform
cost-center=engineering
```

Recommended Property Categories

Category	Example Properties	Purpose
Environment	env=prod , env=staging , env=dev	Distinguish environments
Owner	team=platform , team=checkout	Identify responsible team
Application	app=ecommerce , app=mobile-api	Group by application
Cost Center	cost-center=marketing	Financial allocation
Tier	tier=frontend , tier=backend	Architecture layer

Cloud Provider Tags

Cloud tags are automatically imported when using cloud integrations:

Cloud	Tag Format	DQL Field
AWS	AWS resource tags	<code>aws.tag.*</code>
Azure	Azure resource tags	<code>azure.tag.*</code>
GCP	GCP labels	<code>gcp.label.*</code>

AWS Tag Example:

If your EC2 instance has tag `Environment=Production`, it appears as `aws.tag.Environment` in Dynatrace.

Kubernetes Labels

K8s labels are automatically available for container workloads:

K8s Metadata	DQL Field
Namespace	<code>k8s.namespace.name</code>
Deployment	<code>k8s.deployment.name</code>
Pod labels	<code>k8s.pod.labels.*</code>
Node labels	<code>k8s.node.labels.*</code>

4. Segments for Data Filtering

Segments provide DQL-based filtering to create focused views of your data.

Location: Observe and explore → Segments

What are Segments?

Segments are reusable DQL filters that:

- Filter data in Notebooks, Dashboards, and Apps
- Can be applied as default context
- Are shareable across the organization

Creating a Segment

1. Go to Observe and explore → Segments
2. Click "Create segment"
3. Define your DQL filter:

dql dt.entity.host.properties.env == "production"
4. Name your segment (e.g., "Production Environment")
5. Save

Segment Use Cases

Use Case	Segment Filter	Purpose
Environment	properties.env == "prod"	Focus on production
Team	properties.team == "checkout"	Team-specific view
Application	service.name contains "payment"	Application focus
Region	aws.tag.Region == "us-east-1"	Geographic filtering

Segments vs Legacy Management Zones

Feature	Segments	Management Zones (Legacy)
Filter basis	DQL expressions	Rule-based matching
Data types	All Grail data	Entities only
Flexibility	Highly flexible	Limited rule types
Access control	Use Policies instead	Built-in
Modern platform	✅ Recommended	⚠️ Being deprecated

Segment Best Practices

Practice	Why
Use host properties	Consistent filtering
Name clearly	Prod-Checkout-Team not Segment1

Practice	Why
Document purpose	Add description
Test filters	Verify expected data

5. Naming Conventions

Consistent naming makes entities discoverable and filtering effective.

Host Naming

Set meaningful host names that encode key information:

Pattern	Example	Components
{env}-{tier}-{seq}	prod-web-01	Environment, tier, sequence
{region}-{app}-{role}	us-east-ecom-api	Region, app, role
{team}-{service}-{id}	checkout-cart-a1b2	Team, service, unique ID

Host Naming via OneAgent

You can set a custom display name during installation:

```
sudo /bin/sh Dynatrace-OneAgent.sh --set-host-name="prod-web-01"
```

Naming Principles

Principle	Good	Bad
Descriptive	payment-service	svc-001
Consistent	prod-web-01 , prod-web-02	prod-web-01 , Web Server 2
Parseable	us-east-prod-checkout	USEastProdCheckout
Unique	Include environment/region	Generic names

Property Naming Standards

For host properties, use consistent naming:

Standard	Example	Why
Lowercase	env=prod not ENV=PROD	Consistency
Hyphen separated	cost-center=eng	Readability
Short keys	env not environment	Query simplicity
Consistent values	Always prod not sometimes production	Filtering works

6. Querying by Tags and Properties

Use host properties and cloud tags in DQL queries to filter and group data.

```
// Find hosts by name pattern
fetch dt.entity.host
| filter contains(entity.name, "prod")
| fields entity.name
| limit 20

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | filter contains(name, "prod")
// | fields name
// | limit 20
```

```
// Count hosts by operating system type
fetch dt.entity.host
| summarize host_count = count(), by: {osType}
| sort host_count desc
```

```
// Find services by name pattern
fetch dt.entity.service
| filter contains(entity.name, "checkout")
| fields entity.name, serviceType
| limit 20

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | filter contains(name, "checkout")
// | fields name, serviceType
// | limit 20
```

```
// Query logs filtered by host group (Kubernetes)
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize log_count = count(), by: {k8s.namespace.name}
| sort log_count desc
| limit 20
```

```
// Query spans by service name pattern
fetch spans, from: now() - 1h
| filter span.kind == "server"
| filter contains(service.name, "payment")
| summarize request_count = count(), by: {service.name}
| sort request_count desc
| limit 20
```

```
// Find hosts by name pattern for environment identification
fetch dt.entity.host
| filter not(contains(entity.name, "prod"))
    and not(contains(entity.name, "staging"))
    and not(contains(entity.name, "dev"))
| fields entity.name
| limit 20
```

7. Next Steps

With organization in place:

1. **ONBRD-07: Understanding Your Data** - Explore what Dynatrace discovered
2. Define host properties for your environment

3. Create segments for team-specific views
4. Document your naming conventions

Organization Checklist

- [] Host property strategy documented
 - [] Properties set on OneAgent installations
 - [] Cloud tags verified (if using cloud providers)
 - [] Segments created for common filters
 - [] Naming conventions established
 - [] Access control configured via Policies (see ONBRD-02)
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Summary

In this notebook, you learned:

- Why organization matters for scalability
 - The modern "tag at source" approach
 - How to use host properties and cloud tags
 - How to create and use Segments for filtering
 - Naming convention best practices
 - How to query by properties and attributes
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References

- [Host Properties](#)
 - [Segments](#)
 - [Cloud Tags](#)
 - [Kubernetes Labels](#)
 - [DQL Reference](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.